

4 (a) ($p =$) mv C1

$$\Delta p (= -6.64 \times 10^{-27} \times 1250 - 6.64 \times 10^{-27} \times 1250) = 1.66 \times 10^{-23} \text{ N s}$$

A1 [2]

(b) (i) molecule collides with wall/container **and** there is a change in momentum B1

change in momentum / time is force or $\Delta p = Ft$ B1

many/all/sum of molecular collisions over surface/area of container produces pressure B1 [3]

(ii) more collisions per unit time so greater pressure B1 [1]